

Introduction

QPSK (quadrature phase-shift keying) is a digital modulation scheme that encodes 2 bits in each symbol. Each of the in-phase (I) and quadrature (Q) components in a symbol can be non-inverted (+1) or inverted (-1), with a constant amplitude and carrier frequency. This yields 4 possible complex symbols: $1+j$, $1-j$, $-1+j$, and $-1-j$. Because all of these symbols have the same amplitude of $\sqrt{2}$, QPSK is said to be *amplitude independent*. Each symbol corresponds to a two-bit combination: one for the sign of the in-phase component, and another for the quadrature sign.

<u>Bit position</u>	<u>Definition</u>	<u>Value if bit is 0</u>	<u>Value if bit is 1</u>
1	In-phase sign	positive	negative
0 (LSB)	Quadrature sign	positive	negative

Table 1. Bit scheme for QPSK

Project Overview

The goal of this project is to design a wireless QPSK transceiver system, with a 12.4 GHz carrier, a 3 Gbps data rate (meaning a symbol rate of 1.5 GHz since QPSK symbols contain 2 bits), and transmission power of 11.5-14.5 dBm. To ensure reliability, sources of error such as noise or intermodulation products should be limited. The receiver must have at least 30 dB SNR and 3.6 dB noise figure, while the outputs of both transmitter and receiver must have a 43 dB signal-to-spurious ratio.

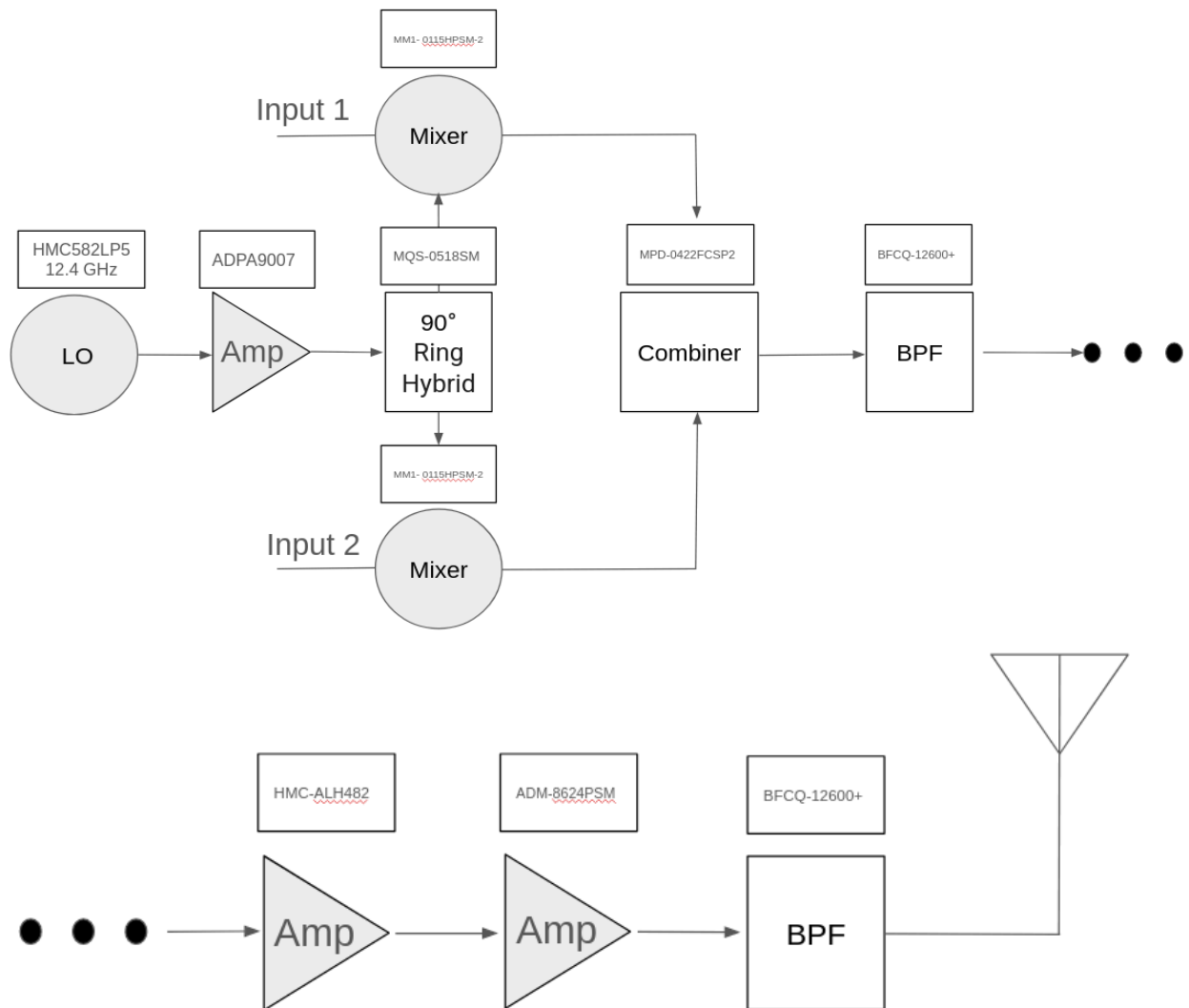
Transmitter

Figure 1. Complete Block Diagram of the Designed Transmitter

The transmitter is designed to send a $0 \text{ GHz} \pm 1.5 \text{ GHz}$ QPSK digital signal, which is first frequency-shifted to prevent overlap between components. This shifted baseband signal is then upconverted using a 12.4 GHz local oscillator to produce a modulated signal centered at 12.4 GHz. The target transmit power at the output is set to approximately $13 \text{ dBm} \pm 1.5 \text{ dBm}$.

A. Local Oscillator Carrier Chain

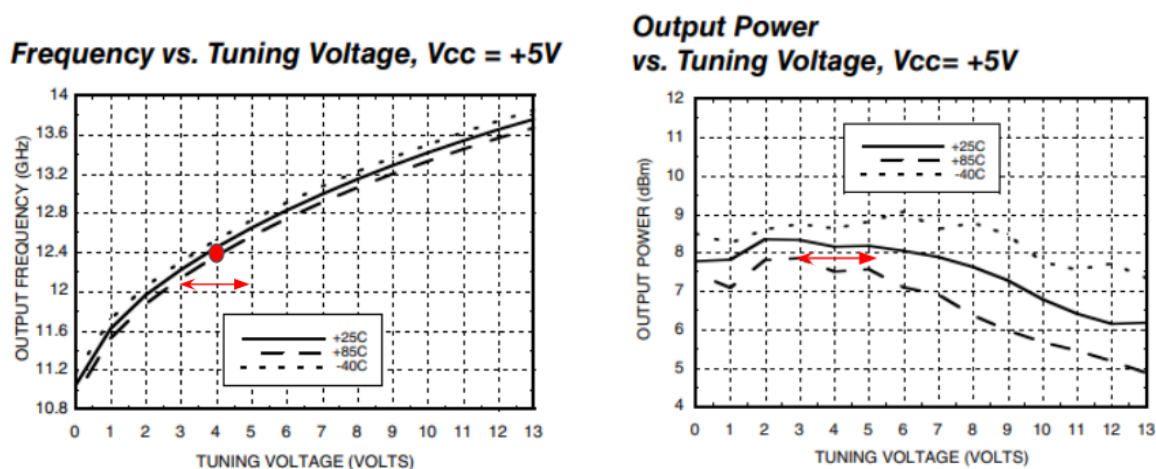


Figure 2. Frequency vs. Tuning Voltage and Output Power vs. Tuning Voltage of the Local Oscillator HMC529LP5 ($V_{cc} = +5\text{ V}$)

The 12.4 GHz local oscillator signal is generated using the HMC529LP5 and initially amplified by a low-noise amplifier (ADPA9007) to ensure it meets the required drive level for the mixers in the upconversion chain. As shown in Figure 2, the output power of the local oscillator is estimated to be a minimum of 8 dBm, a typical value of 8.2 dBm, and a maximum of 8.5 dBm.

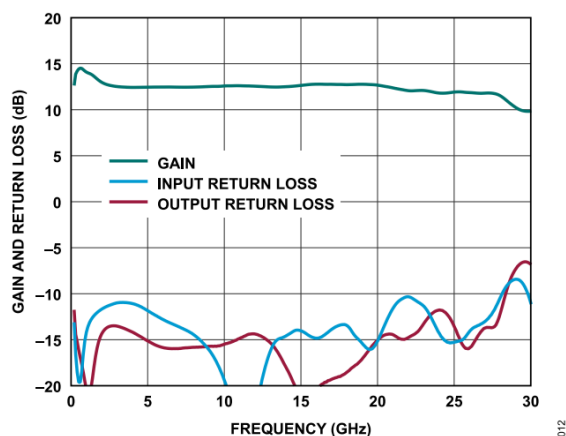


Figure 3. Gain and Return Loss vs. Frequency (200 MHz to 30 GHz) for the low-noise amplifier (ADPA9007), with $V_{DD} = 15\text{ V}$ and $I_{DQ} = 500\text{ mA}$

As described in Figure 3, the amplifier used has a gain of a minimum of 12.3 dB, a typical value of 12.5 dB, and a maximum of 12.7 dB, with a P1dB of a minimum of

31 dBm and a typical value of 33 dBm. This boosts the LO signal power to a minimum of 20.3 dBm, a typical value of 20.7 dBm, and a maximum of 21.2 dBm.

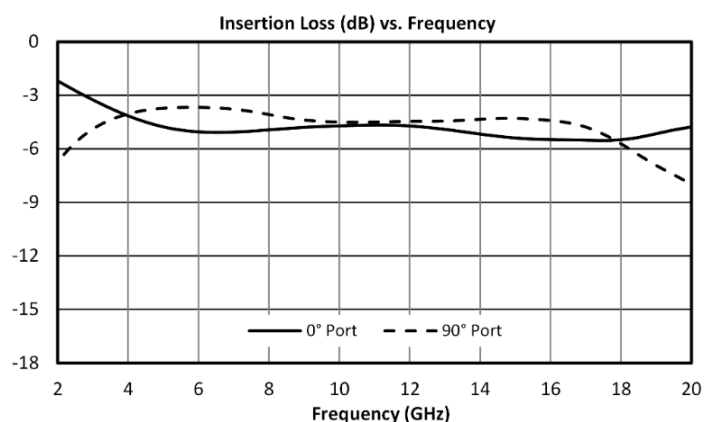


Figure 4. Insertion Loss (dB) vs. Frequency for the 90° ring hybrid (MQS-0518SM)

Following amplification, the LO signal is split using a 90° ring hybrid (MQS-0518SM), with typical performance parameters as specified in its datasheet. As shown in Figure 4, the hybrid exhibits a typical insertion loss of 4.4 dB, resulting in output power levels of a minimum of 15.9 dBm, a typical value of 16.3 dBm, and a maximum of 16.8 dBm. This enables accurate I/Q modulation and ensures that the two orthogonal bitstreams remain properly isolated, preventing interference during the upconversion process.

B. Main Chain

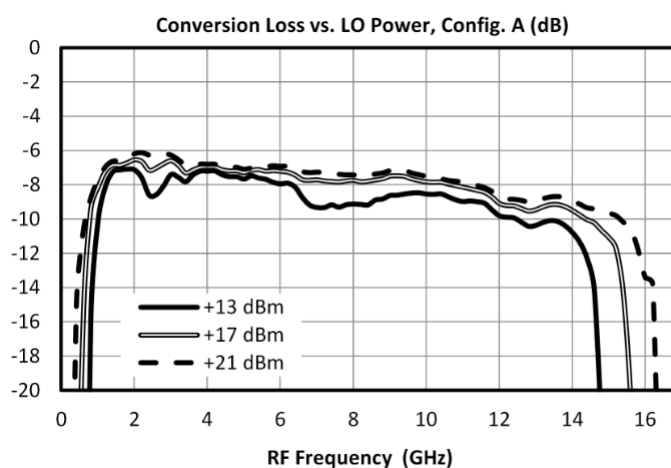


Figure 5. Conversion Loss vs. LO Power for the Mixer (MM1-0115HPSM-2)

Then, the QPSK signals are upconverted using 12.4 GHz sinusoidal signals via upconversion mixers (MM1-0115HPSM-2), resulting in signals centered at 12.9 GHz $\pm$ 1.5 GHz. Mathematically, this process can be represented as:

$$Upconverted_I(t) = I(t) * \cos(2 * \pi * f_{LO} * t)$$

$$Upconverted_Q(t) = Q(t) * \sin(2 * \pi * f_{LO} * t)$$

where $f_{LO} = 12.4$ GHz, and $I(t)$, $Q(t)$ are the in-phase and quadrature baseband signals, respectively. After upconversion, each signal has a power level of a minimum of -10.5 dBm, a typical value of -9.0 dBm, and a maximum of -7.0 dBm.

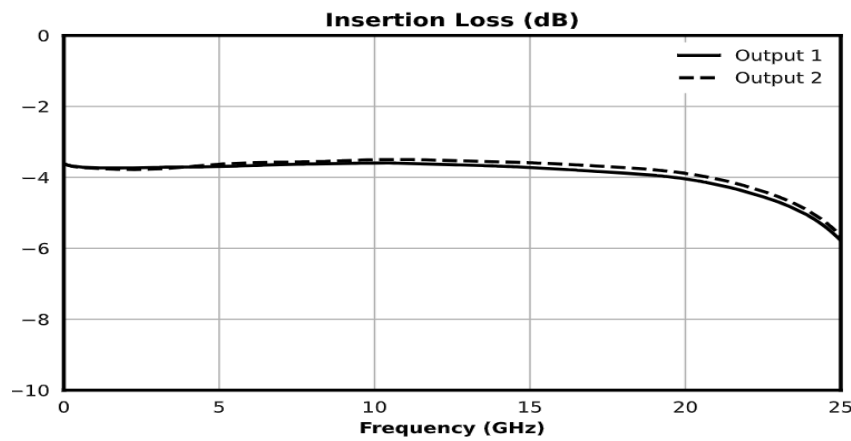


Figure 6. Insertion Loss vs. Frequency for the Wilkinson Combiner (MPD-0422FCSP2)

These upconverted signals are then combined into a single, non-overlapping output using a Wilkinson combiner (MPD-0422FCSP2). From Figure 6, the excess loss can be determined by measuring how much additional loss exists beyond the expected -3 dB. Taking into account this excess loss and the 3 dB gain theoretically achieved when combining two coherent signals, the output power is estimated to be a minimum of -8.2 dBm, a typical value of -6.5 dBm, and a maximum of -4.4 dBm.

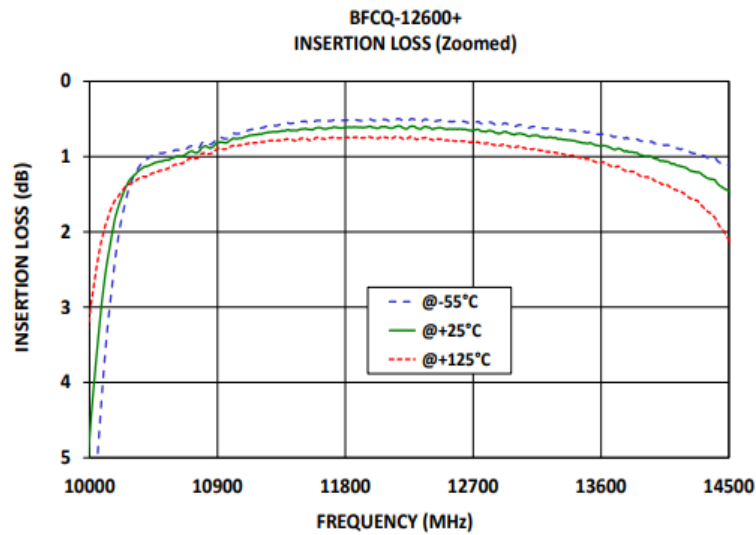


Figure 7. Insertion Loss vs. Frequency for the Bandpass Filter (BFCQ-12600+)

The combined signal is passed through a bandpass filter (BFCQ-12600+) to suppress the image frequency around $0 \text{ GHz} \pm 1.5 \text{ GHz}$ generated during the mixing process. The filter has a passband ranging from 10.7 GHz to 14.2 GHz, which is sufficient to pass the desired signal while effectively attenuating the image frequency in the 0–1.5 GHz range. It exhibits an insertion loss of a minimum of 0.5 dB, a typical value of 0.7 dB, and a maximum of 1.0 dB, resulting in output signal power levels of a minimum of -9.2 dBm , a typical value of -7.2 dBm , and a maximum of -4.9 dBm .

Linear Gain vs. Frequency

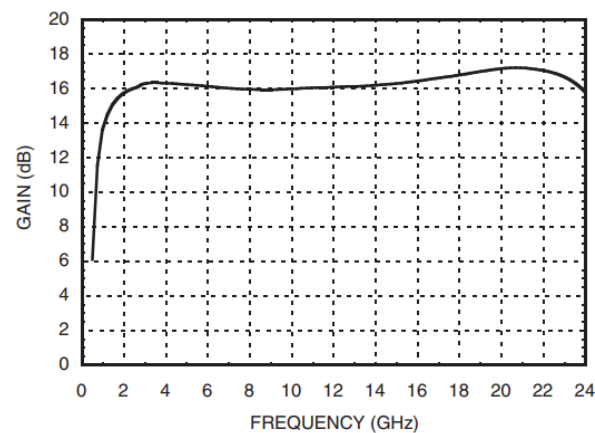


Figure 8. Linear Gain vs. Frequency for the Amplifier (HMC-ALH482)

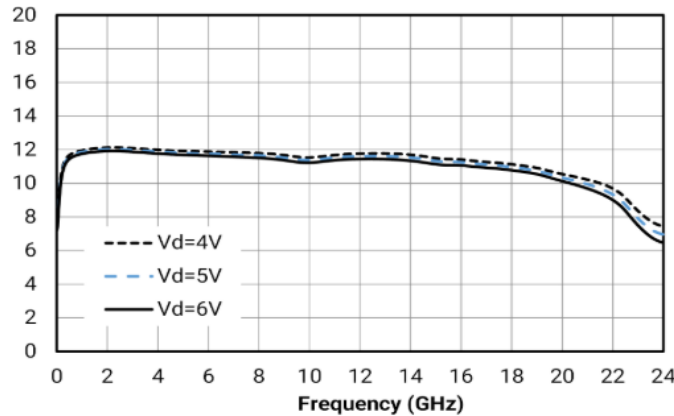


Figure 9. Small Signal Gain vs. Frequency for the Amplifier (HMC-ALH482) with $I_{DQ} = 40$ mA

Next, the signal is amplified by two low-noise amplifiers. The first amplifier (HMC-ALH482) provides a gain of a minimum of 16.0 dB, a typical value of 16.0 dB, and a maximum of 16.2 dB, resulting in output power levels of a minimum of 6.8 dBm, a typical value of 8.8 dBm, and a maximum of 11.3 dBm. The second amplifier offers an additional gain of a minimum of 11.1 dB, a typical value of 11.4 dB, and a maximum of 11.7 dB, with a tunable P1dB range from 13.0 to 13.5 dBm. According to its datasheet, the amplifier's saturation point is specified at 14.5 dBm, and as a result, the final amplification stage produces compressed output power levels of a minimum of 13.5 dBm, a typical value of 14.0 dBm, and a maximum of 14.5 dBm. However, this level of amplification can increase the presence of spurious harmonics originating from the mixers.

To mitigate spurious harmonics generated during amplification, the same bandpass filter (BFCQ-12600+) is applied again at the output stage, just before the signal is transmitted through an ideal antenna. As previously noted, the bandpass filter introduces an insertion loss of a minimum of 0.5 dB, a typical value of 0.7 dB, and a maximum of 0.8 dB. After filtering, the final transmit power is measured to be a minimum of 12.7 dBm, a typical value of 13.3 dBm, and a maximum of 14.0 dBm, which satisfies the target transmit power range of a minimum of 11.5 dBm, a typical value of 13.0 dBm, and a maximum of 14.5 dBm.

Receiver

Receiver (Data Chain)

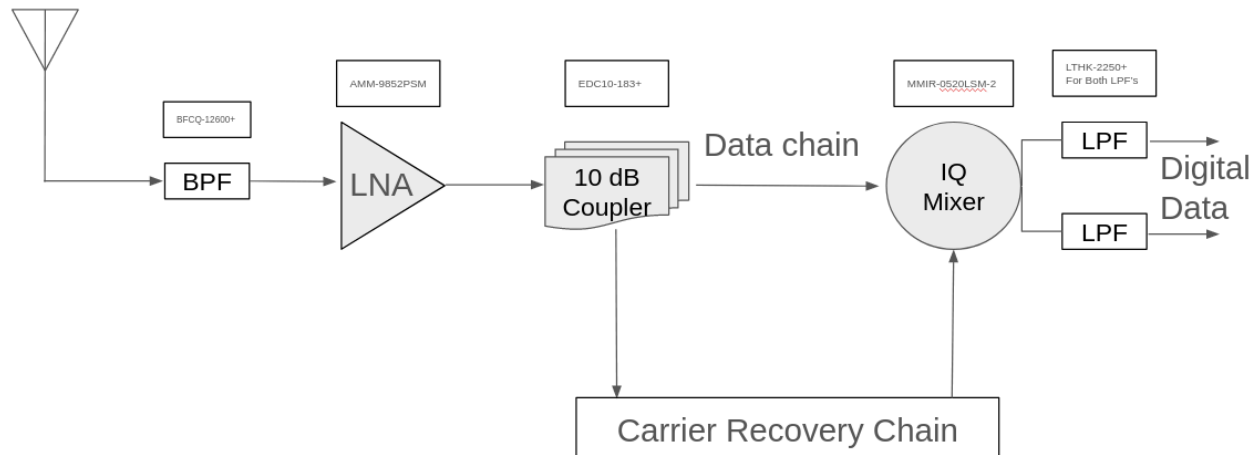


Figure 10. Overall Block Diagram of the Receiver

Once the modulated signal has been transmitted, the receiver has to demodulate it back to the original baseband data. Here, we assume the transmission channel loss is 30dB. The first step of this process is an anti-imaging bandpass filter (BFCQ-12600+), which removes any higher-frequency spurs that would downconvert over the signal band. This filtered signal is then amplified with an LNA (AMM-9852PSM) and fed through a 10 dB coupler (EDC10-183+).

The purpose of the coupler is to avoid a direct junction. (Transmission line junctions cause reflections and standing waves, which can corrupt the phase. Because QPSK encodes all data in the phase, this effect would destroy the signal.) The through port on the coupler feeds into the carrier recovery chain, which will recover the original carrier. The signal itself is taken from the coupled port and fed into an IQ mixer (MMIR-0520LSM-2). Using the newly recovered carrier as an LO, the data is demodulated by the mixer and low-passed (LTHK-2250+) to output the original bitstream.

Receiver (Carrier Recovery Chain)

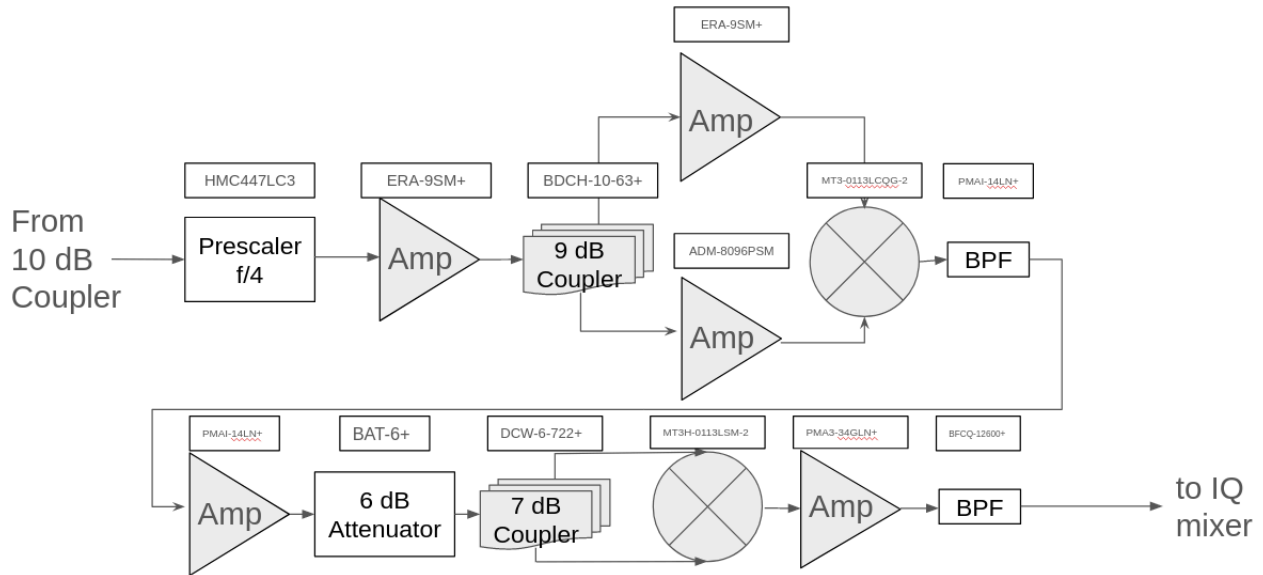


Figure 11. Overall Block Diagram of the Carrier Recovery Chain

The purpose of the carrier recovery chain is to produce a sinusoidal voltage waveform with a constant phase, fulfilling the purpose of a local oscillator signal. The output of this chain is to be fed into a mixer with the received data for demodulation. Because the data signal is a sinusoidal waveform with varying phases, the task of the carrier recovery chain is to eliminate the phase information in the data signal.

At every point in time, the data signal is a waveform whose voltage satisfies the following equation:

$$v_{in}(t) = a_0 \cos(2\pi ft + \phi)$$

Here, ϕ is either $\pi/4$, $3\pi/4$, $5\pi/4$, or $7\pi/4$ depending on the value of the transmitted bits. When multiplying this signal with itself, the resulting waveform becomes:

$$v_{in}^2(t) = \frac{a_0}{2} + \frac{a_0}{2} \cos(4\pi ft + 2\phi)$$

High-pass filtering the signal to remove the DC component then mixing the signal with itself again yields the following waveform:

$$v_1^2(t) = \frac{a_0}{4} + \frac{a_0}{4} \cos(8\pi ft + 4\phi)$$

High-pass filtering this signal to remove the DC component yields the waveform:

$$v_2(t) = \frac{a_0}{4} \cos(8\pi ft + 4\phi)$$

This waveform is a sinusoid with one of four phases, but by examining each case for the value of ϕ , it can be seen that all four of these phases are actually the same:

$$\phi = \frac{\pi}{4} \Rightarrow 4\phi = \pi \Rightarrow v_2(t) = \frac{a_0}{4} \cos(8\pi ft + \pi) = -\frac{a_0}{4} \cos(8\pi ft)$$

$$\phi = \frac{3\pi}{4} \Rightarrow 4\phi = 3\pi \Rightarrow v_2(t) = \frac{a_0}{4} \cos(8\pi ft + 3\pi) = \frac{a_0}{4} \cos(8\pi ft + 3\pi) = -\frac{a_0}{4} \cos(8\pi ft)$$

$$\phi = \frac{5\pi}{4} \Rightarrow 4\phi = 5\pi \Rightarrow v_2(t) = \frac{a_0}{4} \cos(8\pi ft + 5\pi) = \frac{a_0}{4} \cos(8\pi ft + 5\pi) = -\frac{a_0}{4} \cos(8\pi ft)$$

$$\phi = \frac{7\pi}{4} \Rightarrow 4\phi = 7\pi \Rightarrow v_2(t) = \frac{a_0}{4} \cos(8\pi ft + 7\pi) = \frac{a_0}{4} \cos(8\pi ft + 7\pi) = -\frac{a_0}{4} \cos(8\pi ft)$$

We are then left with a sinusoid whose phase completely matches that of the local oscillator at the transmitter side. To convert the frequency of the output waveform to an input waveform, a prescaler that divides the frequency of its input by 4 in its output is incorporated into the recovery chain. Here, the prescaler (HMC447LC3) is placed at the beginning of the chain to drastically reduce operating frequencies, lowering the necessary quality of parts and consequently, cost. We are then left with the negative of the local oscillator signal, hence why an input bit of 1 at the transmitter side corresponds to an output bit of 0 at the receiver side.

In order to mix the output waveform with itself, the signal must first be divided to control the amount of power going into both ports of the mixer. For this purpose, a 9dB coupler (BDCH-10-63+) is placed after the prescaler to divide the power to the two ports on the mixer. Each port of the mixer also requires a minimum amount of power delivered to it, and the -4 dBm output of the prescaler is insufficient to power both ports, especially when divided by the coupler. It is for these reasons that three low-noise amplifiers (ERA-9SM+ before the coupler and after the coupler on the main path, ADM-8096PSM after the coupler on the coupled path) are placed to amplify the prescaler output to levels required by the mixer (MT3-0113LCQG).

The mixed signal is then filtered to remove the DC component as discussed prior, using a bandpass filter (MFBP-00104PSM) in place of the high pass filter. The output level of this signal is too low for the second mixer, so another low-noise amplifier (PMA1-14LN+) is used to boost this output with an attenuator (BAT-6+) placed before it to prevent the amplifier from going into compression. The power is then divided between the two ports by a 7 dB coupler (DCW-6-722+), whose main and coupled outputs are directly delivered to the IF and LO ports of the second mixer (MT3-0113LSM).

The output of the second mixer is then amplified by a low-noise amplifier (PMA3-34GLN+) then filtered to remove the DC component with a bandpass filter (BFCQ-12600+). The output of this bandpass filter is a sine wave whose phase aligns with that of the local oscillator, as is the objective of the carrier recovery chain.

Calculations

Receiver Noise Figure

Receiver noise figure is calculated by determining the cascaded noise figure of the main chain on the receiver side. The following equation was used to determine the cascaded noise figure:

$$F_{Receiver} = F_{BPF} + \frac{F_{LNA} - 1}{G_{BPF}} + \frac{F_{Coupler} - 1}{G_{BPF} * G_{LNA}} + \frac{F_{IQ Mixer} - 1}{G_{BPF} * G_{LNA} * G_{Coupler}} + \frac{F_{LPF} - 1}{G_{BPF} * G_{LNA} * G_{Coupler} * G_{IQ Mixer}}$$

Maximum noise figure and minimum gain values were assumed for the low-noise amplifier and the IQ mixer to account for worst-case conditions. Typical values were used for the remaining components, including the bandpass filter, coupler, and low-pass filter. Table 2 summarizes the individual noise contributions of each component. The last two rows in the table show the cumulative noise figure after each stage in the receiver chain. The components are listed in the order they appear in the system—from the receiving antenna (left) to the digital block inputs (right).

	Band Pass Filter	LNA	Coupler	Mixer	Low Pass Filter
Noise Figure (dB)	0.7	1.8	1.3	10	0.7
Noise Figure (linear)	1.17	1.51	1.35	10.00	1.17
Gain (dB)	-0.7	17	-1.3	-10	-0.7
Gain (linear)	0.85	50.12	0.74	0.10	0.85
Cascaded Noise Figure (dB)	0.70	2.50	2.52	3.16	3.28
Cascaded Noise Figure (linear)	1.17	1.78	1.79	2.07	2.13

Table 2. Receiver Noise Figure

Output SNR

Output SNR is calculated by finding the input SNR and then multiplying the noise figure of the overall system. Here, the input is taken to be devoid of noise aside from the theoretical minimum noise floor of -174 dBc/Hz. Because the input is centered at DC and has a bandwidth of 1.5 GHz, the maximum frequency at the input is 1.5 GHz. The noise level at the input can then be found with the following calculation:

$$N_{input} = -174 \left(\frac{dBm}{Hz} \right) + 10 * \log_{10}(1.5 * 10^9) = -82.2 dBc$$

Taking the input signal power level to be $0 \pm 1\text{ dBm}$, the input SNR is then found by subtracting the noise level.

$$SNR_{in} = P_{in}(\text{dBm}) - N_{floor}(\text{dBm}) = (81.2, 82.2, 83.2) \text{ dB}$$

The system consists of a transmitter, a channel link, and a receiver, all of which have separate noise figures and gains that impact the SNR (with the exception of the receiver gain). The noise figure of the receiver was calculated in the previous section. The channel is assumed to be ideal, attenuating the transmitted signal by 30 dB. The gain of the channel is then -30 dB, meaning the noise figure of the channel is 30 dB. The gain of the transmitter side can be found by subtracting the input power from the output power and was also calculated in a previous section. What remains to be calculated, then, is the transmitter noise figure.

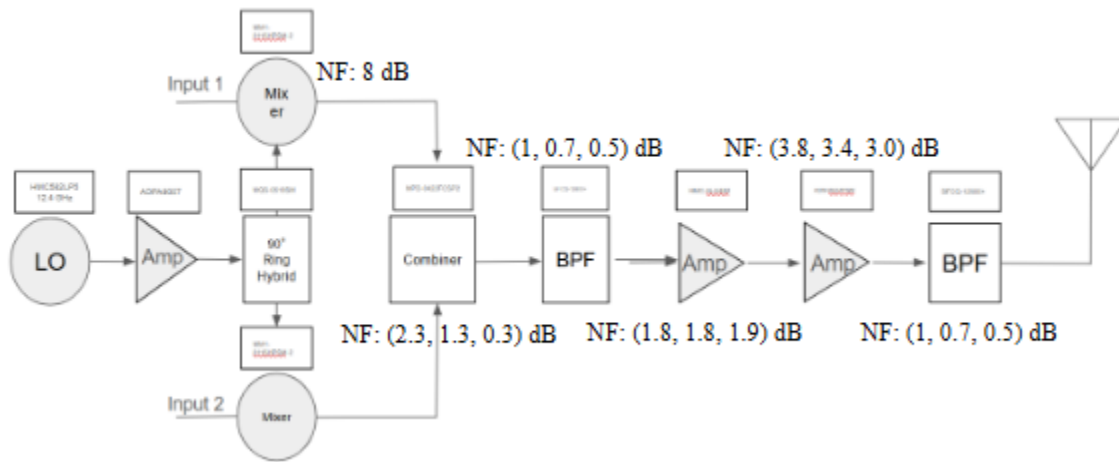


Figure 12. Full Transmitter Diagram with Each Component That Processes the Data Signal Labelled with Its Respective Noise Figure

The gain of each component was discussed in a previous section, so the noise figure of each part remains to be explained. The noise figure of the mixer is taken to be the typical noise figure listed in the datasheet, which is 8 dB. No other noise figure is listed save for that of a different configuration. The combiner and the bandpass filters are passive components, meaning their noise figures are the inverse of their gains, which were found in a previous section. The datasheet for the first amplifier provides the following graph for its noise figure with respect to frequency.

Noise Figure vs. Frequency

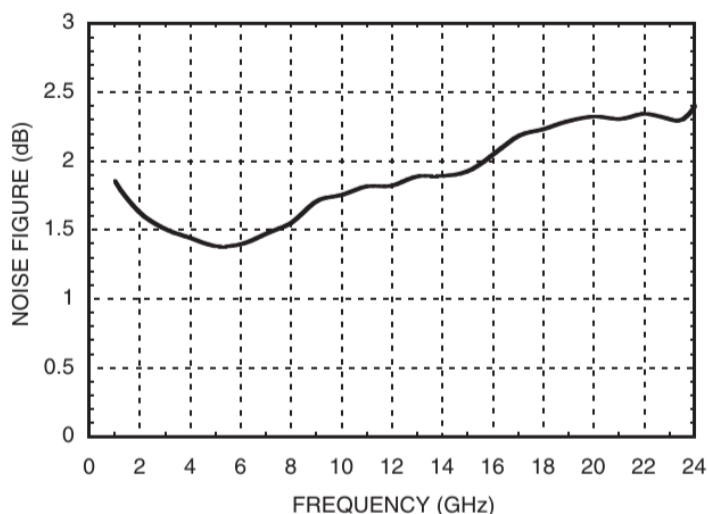


Figure 13. Typical Noise Figure Curve for the HMC-ALH482 Low Noise Amplifier

Within the operating frequency range of 10.9-13.9 GHz, the curve is relatively flat, ranging from 1.8 dB to 1.9 dB. The noise figure of the first amplifier is then taken to be 1.9 dB in the worst case and 1.8 dB in the best and typical cases.

The second amplifier provides the following graph for its noise figure with respect to frequency.

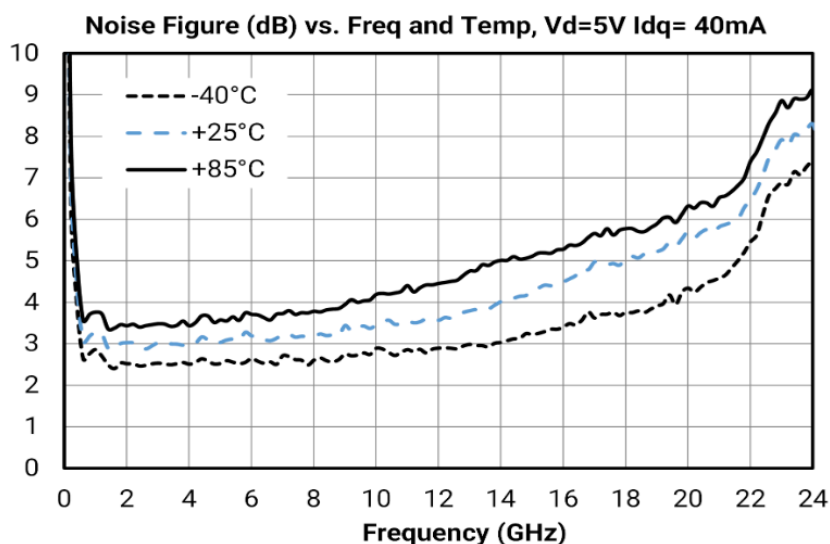


Figure 14. Typical Noise Figure Curve for the ADM-8624PSM Low Noise Amplifier

Again, within the operating frequency range of 10.9-13.9 GHz, the curve is relatively flat, ranging from 3.5 dB to 4 dB. The noise figure of the first amplifier is then taken to be 3.5 dB in the worst case, and 3.7 dB in the typical case, and 4 dB in the best case.

The noise figure of the transmitter can then be calculated via the following equation:

$$F_{TX} = F_{mixer} + \frac{F_{Coupler}}{G_{mixer}} + \frac{F_{BPF}}{G_{mixer}G_{Coupler}} + \frac{F_{LNA1}}{G_{mixer}G_{coupler}G_{BPF}} + \frac{F_{LNA2}}{G_{mixer}G_{coupler}G_{BPF}G_{LNA1}} + \frac{F_{BPF}}{G_{mixer}G_{coupler}G_{BPF}G_{LNA1}G_{LNA2}}$$

The noise figure of the transmitter is found to be 11.8 dB in the best case, 15.4 dB in a typical case, and 19.4 dB in the worst case. The noise figures and gains of each system are provided in the diagram below:

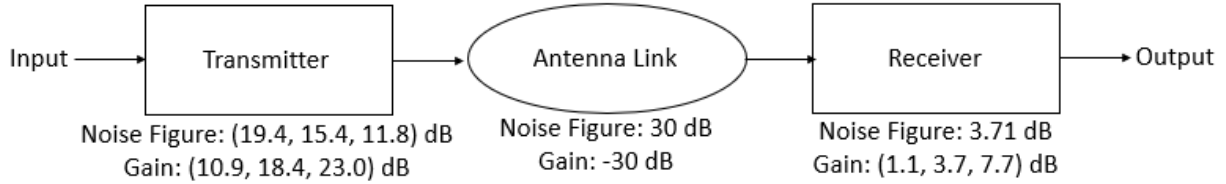


Figure 15. Block Diagram of the Transmitter/Receiver System with the Noise Figure and Gain of Each System

The expression used to calculate the overall noise figure is as follows:

$$F_{Total} = F_{Transmitter} + \frac{F_{Link}}{G_{Transmitter}} + \frac{F_{Receiver}}{G_{transmitter}G_{Link}} = (24.3, 22.2, 19.6) \text{ dB}$$

Subtracting this noise figure from the input SNR then yields the following results:

$$SNR_{out} = SNR_{in}(dB) - F_{Total}(dB) = (56.9, 60.1, 63.6) \text{ dB}$$

The goal here was to achieve a minimum output SNR of 30 dB, which is significantly below the achieved SNR in all cases.

Maximum Output Spurious Signal (Transmitter)

The most significant spurious signals generated in the transmitter side are the IF leakage from the mixer and the 3rd harmonic caused by the second amplifier being in saturation. What follows is an analysis of the levels of these spurious signals at each stage of the transmitter.

The datasheet of the mixers used in the transmitter side provides the following typical curve for RF-IF isolation.

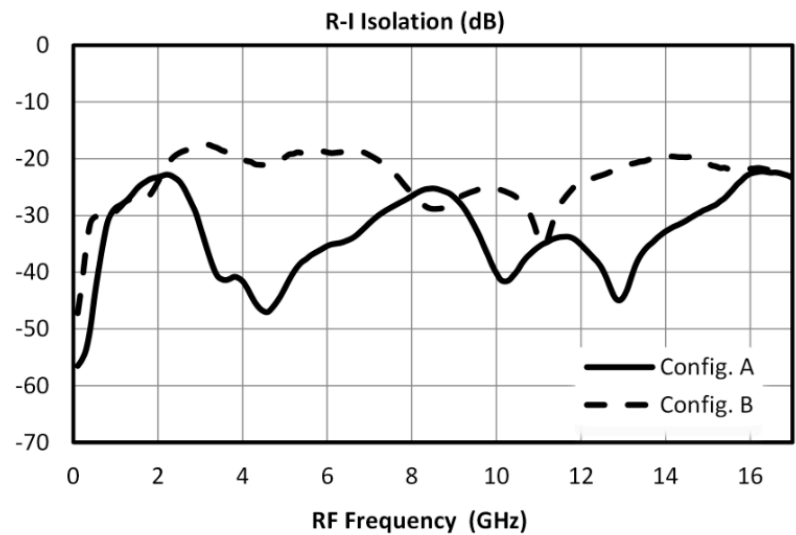


Figure 16. Typical RF-IF Isolation Curve for the MM1-0115HPSM-2 Mixer

Within the RF frequency range of 10.9-13.9 GHz, IF leakage ranges between 33dB and 45 dB. Since the curve is rarely below 40 dB, 40 dB is taken to be the best case RF-IF leakage with 33 dB taken to be the best case leakage and 34 dB taken to be the typical case. The power of this spurious signal at the output of the mixer is then calculated below for each case.

	WORST	TYPICAL	BEST
After Mixer	<i>Desired: -1 dBm - 9.5 dB = -10.5 dBm</i> <i>IF: -1 dBm - 33 dB = -34 dBm</i>	<i>Desired: 0 dBm - 9 dB = -9 dBm</i> <i>IF: 0 dBm - 34 dB = -34 dBm</i>	<i>Desired: 1 dBm - 8 dB = -7 dBm</i> <i>IF: 1 dBm - 40 dB = -39 dBm</i>

The insertion loss curve of the combiner was shown in a previous section to be very flat, varying between 3.7 dB in the best case and 3.4 dB in the worst case. This is when the combiner is used to divide rather than to combine, so the output power of the combiner is 6 dB higher. The power of the spurious signal is raised by 2.3 dB in the best case and by 2.6 dB in the best case. The power of this spurious signal at the output of the combiner is then calculated below.

After Mixer	<i>Desired: -1 dBm - 9.5 dB = -10.5 dBm</i> <i>IF: -1 dBm - 33 dB = -34 dBm</i>	<i>Desired: 0 dBm - 9 dB = -9 dBm</i> <i>IF: 0 dBm - 34 dB = -34 dBm</i>	<i>Desired: 1 dBm - 8 dB = -7 dBm</i> <i>IF: 1 dBm - 40 dB = -39 dBm</i>
After Combiner	<i>Desired: -10.5 dBm + 2.3 dB = -8.2 dBm</i> <i>IF: -32 dBm + 2.6 dB = -29.4 dBm</i>	<i>Desired: -9 dBm + 2.5 dB = -6.5 dBm</i> <i>IF: -34 dBm + 2.5 dB = -31.5 dBm</i>	<i>Desired: -7 dBm + 2.6 dB = -4.4 dBm</i> <i>IF: -39 dBm + 2.3 dB = -36.7 dBm</i>

The typical curve provided in the datasheet for the bandpass filter following the combiner is shown below:

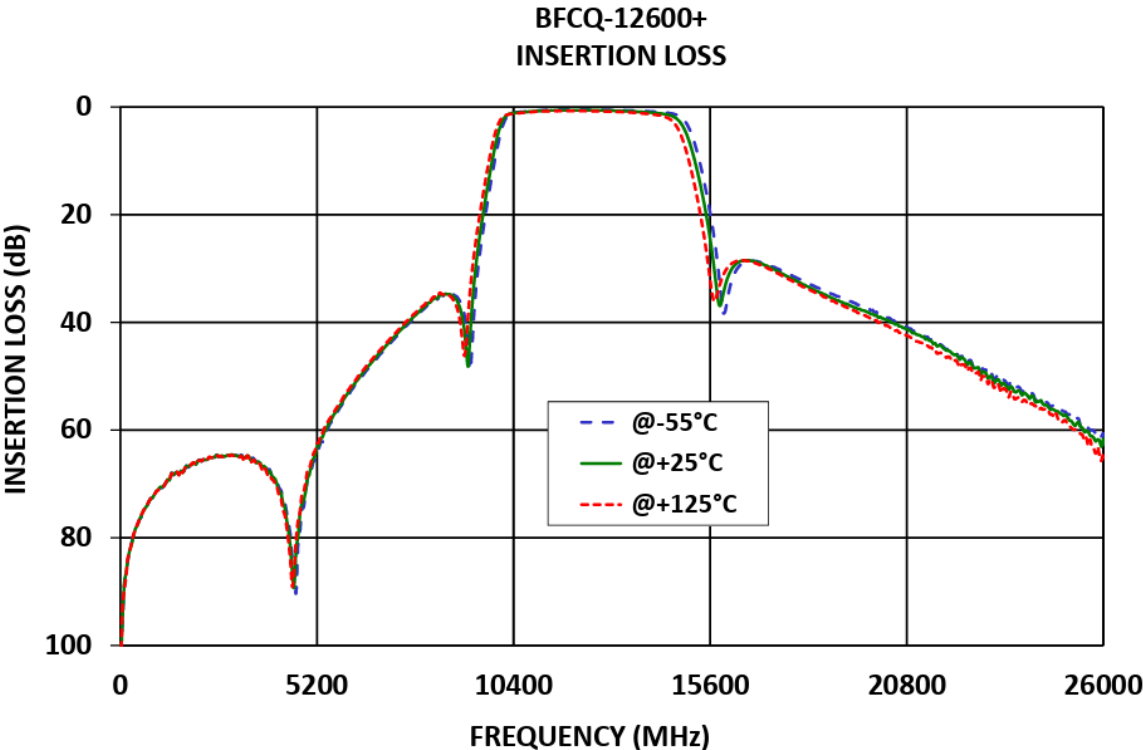


Figure 17. Typical Curve for the Insertion Loss of the BFCQ-12600+ Bandpass Filter

The IF spectrum ranges from 0 to 1.5 GHz, so it can be inferred that attenuation is at least 65 dB in this range. As such, 65 dB attenuation is taken for the worst case, with 80 dB attenuation for the typical case. The curve reaches 100 dB attenuation, but since this is such a small part of the spectrum, best case attenuation is taken to be 90 dB. The power level of the mixer IF leakage spurious signal at the output of the bandpass filter is then as follows:

	WORST	TYPICAL	BEST
After Mixer	<i>Desired:</i> -1 dBm - 9.5 dB = -10.5 dBm <i>IF:</i> -1 dBm - 33 dB = -34 dBm	<i>Desired:</i> 0 dBm - 9 dB = -9 dBm <i>IF:</i> 0 dBm - 34 dB = -34 dBm	<i>Desired:</i> 1 dBm - 8 dB = -7 dBm <i>IF:</i> 1 dBm - 40 dB = -39 dBm
After Combiner	<i>Desired:</i> -10.5 dBm + 2.3 dB = -8.2 dBm <i>IF:</i> -32 dBm + 2.6 dB = -29.4 dBm	<i>Desired:</i> -9 dBm + 2.5 dB = -6.5 dBm <i>IF:</i> -34 dBm + 2.5 dB = -31.5 dBm	<i>Desired:</i> -7 dBm + 2.6 dB = -4.4 dBm <i>IF:</i> -39 dBm + 2.3 dB = -36.7 dBm
After BPF 1	<i>Desired:</i> -8.2 dBm - 1 dB = -9.2 dBm <i>IF:</i> -29.4 dBm - 65 dB = -94.4 dBm	<i>Desired:</i> -6.5 dBm - 0.7 dB = -7.2 dBm <i>IF:</i> -31.5 dBm - 80 dB = -111.5 dBm	<i>Desired:</i> -4.4 dBm - 0.5 dB = -4.9 dBm <i>IF:</i> -36.7 dBm - 90 dB = -129.7 dBm

The typical curve for the gain of the first Low Noise Amplifier is shown below:

Linear Gain vs. Frequency

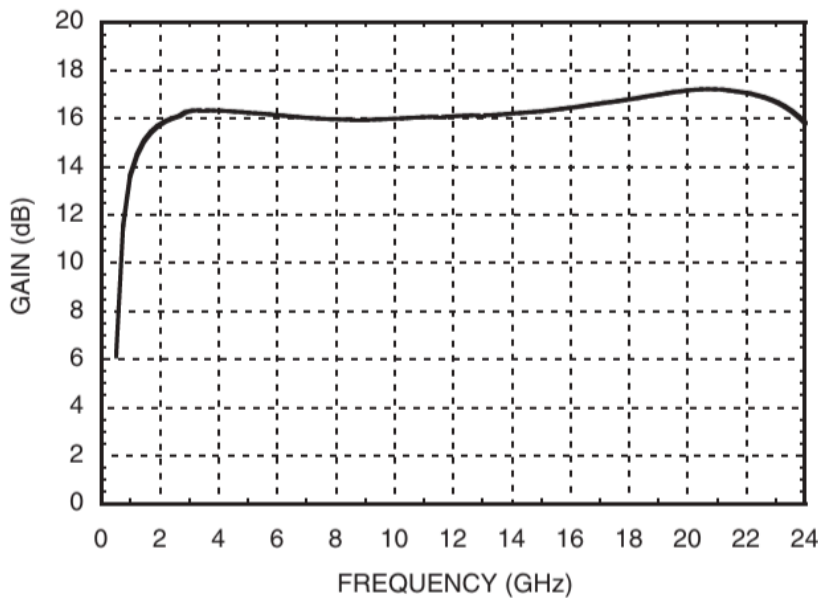


Figure 18: Typical Curve for the Gain of the HMC-ALH482 Amplifier

The gain falls off significantly around the IF leakage frequency range, so a large range of values for the gain is considered here. Gain is taken to be 15 dB in the worst case, 6 dB in the typical case, and 0 dB in the best case. The power level of the spurious signal at the output of the first LNA is then shown below:

After Mixer	<i>Desired:</i> -1 dBm - 9.5 dB = -10.5 dBm <i>IF:</i> -1 dBm - 33 dB = -34 dBm	<i>Desired:</i> 0 dBm - 9 dB = -9 dBm <i>IF:</i> 0 dBm - 34 dB = -34 dBm	<i>Desired:</i> 1 dBm - 8 dB = -7 dBm <i>IF:</i> 1 dBm - 40 dB = -39 dBm
After Combiner	<i>Desired:</i> -10.5 dBm + 2.3 dB = -8.2 dBm <i>IF:</i> -32 dBm + 2.6 dB = -29.4 dBm	<i>Desired:</i> -9 dBm + 2.5 dB = -6.5 dBm <i>IF:</i> -34 dBm + 2.5 dB = -31.5 dBm	<i>Desired:</i> -7 dBm + 2.6 dB = -4.4 dBm <i>IF:</i> -39 dBm + 2.3 dB = -36.7 dBm
After BPF 1	<i>Desired:</i> -8.2 dBm - 1 dB = -9.2 dBm <i>IF:</i> -29.4 dBm - 65 dB = -94.4 dBm	<i>Desired:</i> -6.5 dBm - 0.7 dB = -7.2 dBm <i>IF:</i> -31.5 dBm - 80 dB = -111.5 dBm	<i>Desired:</i> -4.4 dBm - 0.5 dB = -4.9 dBm <i>IF:</i> -36.7 dBm - 90 dB = -129.7 dBm
After LNA 1	<i>Desired:</i> -9.2 dBm + 16 dB = 6.8 dBm <i>IF:</i> -94.4 dBm + 15 dB = -79.4 dBm	<i>Desired:</i> -7.2 dBm + 16 dB = 8.8 dBm <i>IF:</i> -111.5 dBm + 6 dB = -109.5 dBm	<i>Desired:</i> -4.9 dBm + 16.2 dB = 11.3 dBm <i>IF:</i> -129.7 dBm + 0 dB = -129.7 dBm

The typical curve for the gain of the second Low Noise Amplifier is shown below:

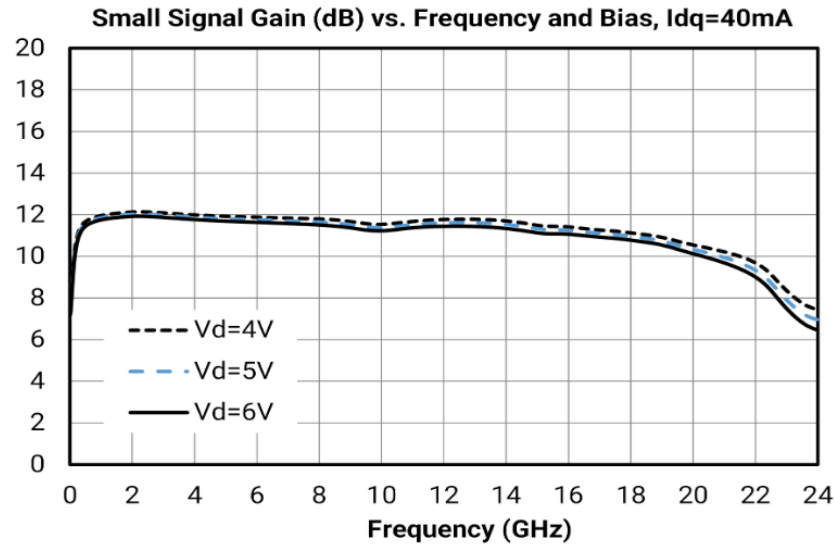


Figure 19. Typical Curve for the Gain of the ADM-8624PSM Amplifier

The gain of this amplifier exhibits similar behavior to that of the first amplifier, so a wide range of values is considered again. Here, 12 dB of gain is taken for the worst case, 9 dB is taken for the typical case, and 0 dB is taken for the best case.

As this amplifier operates in saturation, its output contains more spurious signals – specifically the harmonics of the amplified data signal. For the sake of simplicity, the worst case scenario for the resulting waveform is considered, where the output is a perfect square wave. The most significant harmonic is then the 3rd harmonic, which is at a power level 9.54 dB below the power level of the square wave. The spurious signal power levels at the output of the second LNA are then calculated below:

	WORST	TYPICAL	BEST
After Mixer	Desired: $-1 \text{ dBm} - 9.5 \text{ dB} = -10.5 \text{ dBm}$ IF: $-1 \text{ dBm} - 33 \text{ dB} = -34 \text{ dBm}$	Desired: $0 \text{ dBm} - 9 \text{ dB} = -9 \text{ dBm}$ IF: $0 \text{ dBm} - 34 \text{ dB} = -34 \text{ dBm}$	Desired: $1 \text{ dBm} - 8 \text{ dB} = -7 \text{ dBm}$ IF: $1 \text{ dBm} - 40 \text{ dB} = -39 \text{ dBm}$
After Combiner	Desired: $-10.5 \text{ dBm} + 2.3 \text{ dB} = -8.2 \text{ dBm}$ IF: $-32 \text{ dBm} + 2.6 \text{ dB} = -29.4 \text{ dBm}$	Desired: $-9 \text{ dBm} + 2.5 \text{ dB} = -6.5 \text{ dBm}$ IF: $-34 \text{ dBm} + 2.5 \text{ dB} = -31.5 \text{ dBm}$	Desired: $-7 \text{ dBm} + 2.6 \text{ dB} = -4.4 \text{ dBm}$ IF: $-39 \text{ dBm} + 2.3 \text{ dB} = -36.7 \text{ dBm}$
After BPF 1	Desired: $-8.2 \text{ dBm} - 1 \text{ dB} = -9.2 \text{ dBm}$ IF: $-29.4 \text{ dBm} - 65 \text{ dB} = -94.4 \text{ dBm}$	Desired: $-6.5 \text{ dBm} - 0.7 \text{ dB} = -7.2 \text{ dBm}$ IF: $-31.5 \text{ dBm} - 80 \text{ dB} = -111.5 \text{ dBm}$	Desired: $-4.4 \text{ dBm} - 0.5 \text{ dB} = -4.9 \text{ dBm}$ IF: $-36.7 \text{ dBm} - 90 \text{ dB} = -129.7 \text{ dBm}$
After LNA 1	Desired: $-9.2 \text{ dBm} + 16 \text{ dB} = 6.8 \text{ dBm}$ IF: $-94.4 \text{ dBm} + 15 \text{ dB} = -79.4 \text{ dBm}$	Desired: $-7.2 \text{ dBm} + 16 \text{ dB} = 8.8 \text{ dBm}$ IF: $-111.5 \text{ dBm} + 6 \text{ dB} = -109.5 \text{ dBm}$	Desired: $-4.9 \text{ dBm} + 16.2 \text{ dB} = 11.3 \text{ dBm}$ IF: $-129.7 \text{ dBm} + 0 \text{ dB} = -129.7 \text{ dBm}$
After LNA 2	Desired: 13.5 dBm (1 dB Compression) IF: $-79.4 \text{ dBm} + 12 \text{ dB} = -67.4 \text{ dBm}$ 3rd Harmonic: $13.5 \text{ dBm} - 9.54 \text{ dB} = 3.96 \text{ dBm}$	Desired: 14 dBm (Just below saturation) IF: $-109.5 \text{ dBm} + 9 \text{ dB} = -100.5 \text{ dBm}$ 3rd Harmonic: $14 \text{ dBm} - 9.54 \text{ dB} = 4.46 \text{ dBm}$	Desired: 14.5 dBm (Saturation) IF: $-129.7 \text{ dBm} + 0 \text{ dB} = -129.7 \text{ dBm}$ 3rd Harmonic: $14.5 \text{ dBm} - 9.54 \text{ dB} = 4.96 \text{ dBm}$

At this point, the power level of the third harmonic eclipses that of the IF leakage, so it is apparent that this power level is the highest level spurious output of the transmitter. As such, further analysis concerns only this harmonic, whose bandwidth ranges from 32.7 GHz to 41.7 GHz, triple that of the fundamental harmonic.

Around this frequency range, the attenuation of the BFCQ-12600+ bandpass filter (typical curve shown previously) ranges between 43 dB and 70 dB. As such, the power level at the output of this bandpass filter and therefore the output of the transmitter is calculated below.

	WORST	TYPICAL	BEST
After Mixer	<i>Desired: -1 dBm - 9.5 dB = -10.5 dBm</i> <i>IF: -1 dBm - 33 dB = -34 dBm</i>	<i>Desired: 0 dBm - 9 dB = -9 dBm</i> <i>IF: 0 dBm - 34 dB = -34 dBm</i>	<i>Desired: 1 dBm - 8 dB = -7 dBm</i> <i>IF: 1 dBm - 40 dB = -39 dBm</i>
After Combiner	<i>Desired: -10.5 dBm + 2.3 dB = -8.2 dBm</i> <i>IF: -32 dBm + 2.6 dB = -29.4 dBm</i>	<i>Desired: -9 dBm + 2.5 dB = -6.5 dBm</i> <i>IF: -34 dBm + 2.5 dB = -31.5 dBm</i>	<i>Desired: -7 dBm + 2.6 dB = -4.4 dBm</i> <i>IF: -39 dBm + 2.3 dB = -36.7 dBm</i>
After BPF 1	<i>Desired: -8.2 dBm - 1 dB = -9.2 dBm</i> <i>IF: -29.4 dBm - 65 dB = -94.4 dBm</i>	<i>Desired: -6.5 dBm - 0.7 dB = -7.2 dBm</i> <i>IF: -31.5 dBm - 80 dB = -111.5 dBm</i>	<i>Desired: -4.4 dBm - 0.5 dB = -4.9 dBm</i> <i>IF: -36.7 dBm - 90 dB = -129.7 dBm</i>
After LNA 1	<i>Desired: -9.2 dBm + 16 dB = 6.8 dBm</i> <i>IF: -94.4 dBm + 15 dB = -79.4 dBm</i>	<i>Desired: -7.2 dBm + 16 dB = 8.8 dBm</i> <i>IF: -111.5 dBm + 6 dB = -109.5 dBm</i>	<i>Desired: -4.9 dBm + 16.2 dB = 11.3 dBm</i> <i>IF: -129.7 dBm + 0 dB = -129.7 dBm</i>
After LNA 2	<i>Desired: 13.5 dBm (1 dB Compression)</i> <i>IF: -79.4 dBm + 12 dB = -67.4 dBm</i> <i>3rd Harmonic: 13.5 dBm - 9.54 dB = 3.96 dBm</i>	<i>Desired: 14 dBm (Just below saturation)</i> <i>IF: -109.5 dBm + 9 dB = -100.5 dBm</i> <i>3rd Harmonic: 14 dBm - 9.54 dB = 4.46 dBm</i>	<i>Desired: 14.5 dBm (Saturation)</i> <i>IF: -129.7 dBm + 0 dB = -129.7 dBm</i> <i>3rd Harmonic: 14.5 dBm - 9.54 dB = 4.96 dBm</i>
After BPF 2	<i>Desired: 13.5 dBm - 1 dB = 12.5 dBm</i> <i>3rd Harmonic: 3.96 dBm - 43 dB = -39.04 dBm</i>	<i>Desired: 14 dBm - 0.7 dB = 13.3 dBm</i> <i>3rd Harmonic: 4.46 dBm - 55 dB = -50.54 dBm</i>	<i>Desired: 14.5 dBm - 0.5 dB = 14 dBm</i> <i>3rd Harmonic: 4.96 dBm - 70 dB = -65.04 dBm</i>

With the output powers of the desired input and the maximum output spurious signal shown above, the ratio is calculated.

Maximum Spurious Output	$12.5 \text{ dBm} + 39.04 \text{ dBm} = 51.5 \text{ dBc}$	$13.3 \text{ dBm} + 50.54 \text{ dBm} = 63.8 \text{ dBc}$	$14 \text{ dBm} + 65.04 \text{ dBm} = 79 \text{ dBc}$
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The goal was to achieve a maximum spurious signal output of at most 43 dBc below the fundamental. This ratio is surpassed in all cases.

Maximum Output Spurious Signal (Receiver)

The spurious signals generated from the recovery chain of the receiver were calculated to be too low to drive the mixer at the output. The spurious signals to be considered at the receiver side then all come from the mixer, being the RF leakage signal, the LO leakage signal, and the signal generated from upconversion. Calculations that yields to power levels of the inputs to the mixer are omitted, but the power level of the recovered carrier was calculated to be 9.0 dBm in the worst case (for spurious signals), 8.3 dBm in the typical case, and 7.3 dBm in the best case. The power level of the data signal was

calculated to be 1.0 dBm at most and -3.6 dBm at best, with -1.5 dBm taken as a typical case.

The datasheet for the mixer provides the following typical curves for LO-IF isolation, RF-IF isolation, and conversion loss.

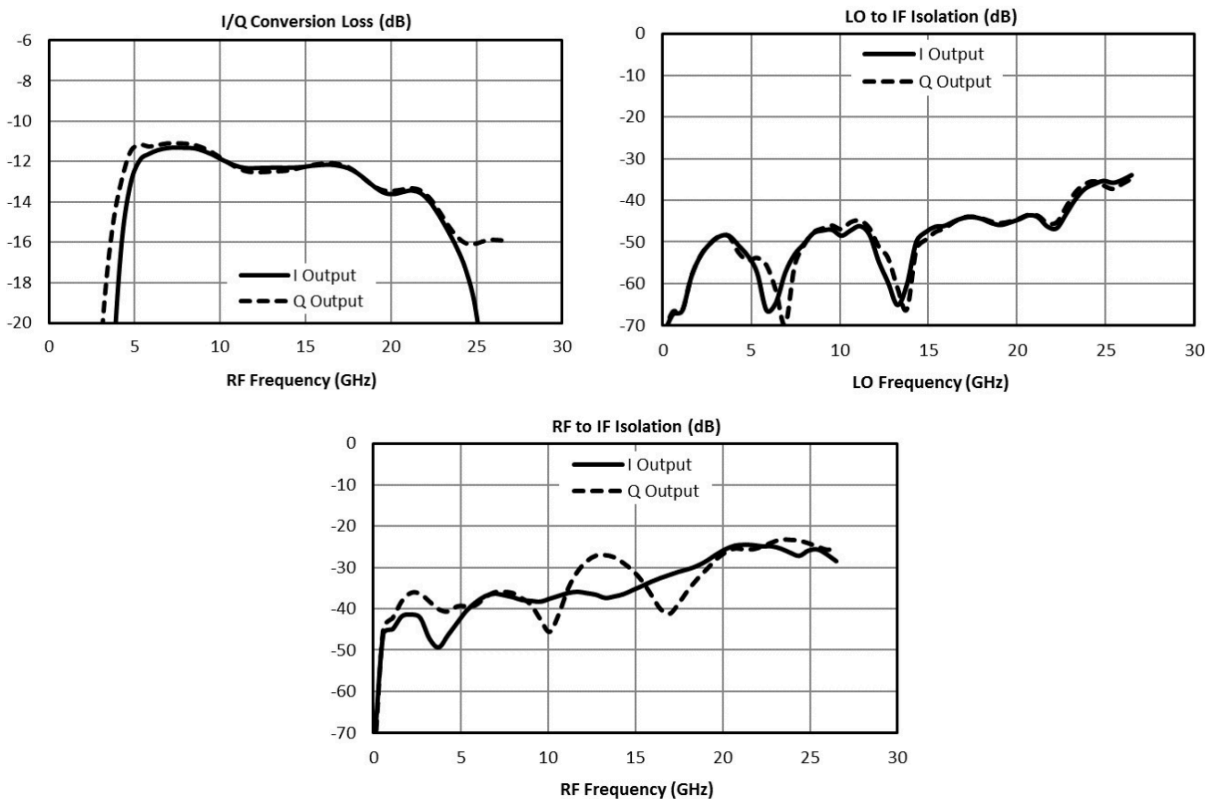


Figure 20. Typical Curves for the Conversion Loss, LO-IF Isolation and RF-IF Isolation of the MMIQ-0520LSM IQ Mixer

Using these typical curves, the following powers of each spurious signal as well as the power of the output signal were calculated as shown below.

	WORST	TYPICAL	BEST
After Mixer	Desired: -3.6 dBm - 10.0 dB = -13.6 dBm RF: -3.6 dBm - 28 dB = -31.6 dBm LO: 9.0 dBm - 45 dB = -36.0 dBm 2 × RF: -3.6 dBm - 9.6 dB = -13.2 dBm	Desired: -1.5 dBm - 9.8 dB = -11.3 dBm RF: -1.5 dBm - 37 dB = -38.5 dBm LO: 8.3 dBm - 49 dB = -57.3 dBm 2 × RF: -1.5 dBm - 9.8 dB = -11.3 dBm	Desired: 1.0 dBm - 9.6 dB = -8.6 dBm RF: 1.0 dBm - 37 dB = -36 dBm LO: 7.3 dBm - 60 dB = -52.7 dBm 2 × RF: 1.0 dBm - 10.0 dB = -9.0 dBm

The typical curve for the insertion loss of the low pass filter placed after each output of the IQ mixer is shown below.

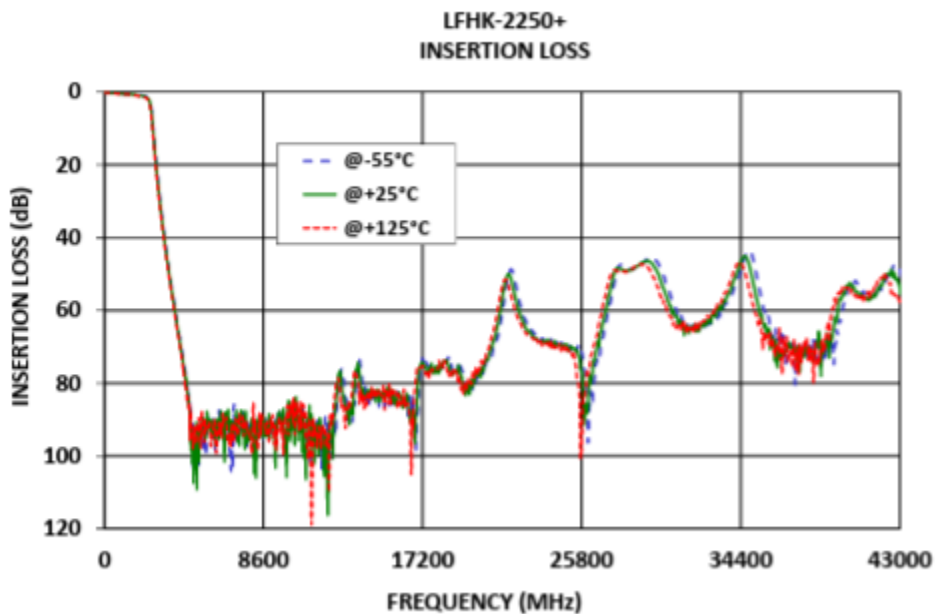


Figure 21. Typical curve for the insertion loss of the LFHK-2250+ low pass filter

The signal levels at the final output of the receiver are then calculated below.

	WORST	TYPICAL	BEST
After Mixer	Desired: -3.6 dBm - 10.0 dB = -13.6 dBm RF: -3.6 dBm - 28 dB = -31.6 dBm LO: 9.0 dBm - 45 dB = -36.0 dBm 2 × RF: -3.6 dBm - 9.6 dB = -13.2 dBm	Desired: -1.5 dBm - 9.8 dB = -11.3 dBm RF: -1.5 dBm - 37 dB = -38.5 dBm LO: 8.3 dBm - 49 dB = -57.3 dBm 2 × RF: -1.5 dBm - 9.8 dB = -11.3 dBm	Desired: 1.0 dBm - 9.6 dB = -8.6 dBm RF: 1.0 dBm - 37 dB = -36 dBm LO: 7.3 dBm - 60 dB = -52.7 dBm 2 × RF: 1.0 dBm - 10.0 dB = -9.0 dBm
After LPF	Desired: -13.6 dBm - 2.3 dB = -15.9 dBm RF: -31.6 dBm - 50 dB = -81.6 dBm LO: -36.0 dBm - 50 dB = -86.0 dBm 2 × RF: -13.2 dBm - 50 dB = -63.2 dBm	Desired: -11.3 dBm - 1.5 dB = -12.8 dBm RF: -38.5 dBm - 68 dB = -106.5 dBm LO: -57.3 dBm - 68 dB = -125.3 dBm 2 × RF: -11.3 dBm - 70 dB = -81.3 dBm	Desired: -8.6 dBm - 0 dB = -8.6 dBm RF: -36 dBm - 100 dB = -136 dBm LO: -52.7 dBm - 100 dB = -152.7 dBm 2 × RF: -9.0 dBm - 80.0 dB = -89.0 dBm

The calculations show that the highest level spurious output at the receiver comes from the upconversion of the mixer and the ratio between the output power of the transmitted data and this spurious signal is calculated below.

Maximum Spurious Output	-15.9 dBm + 63.2 dBm = 47.3 dBc	-12.8 dBm + 81.3 dBm = 68.5 dBc	-8.6 dBm + 89.0 dBm = 80.4 dBc
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The specification to meet here was also a minimum of 43 dBc for this ratio, which is surpassed in all cases.

Phase Error

IQ modulation in the transceiver is performed by the ring hybrid and IQ mixer. Therefore, phase errors in the transmitted symbols can be calculated in terms of the phase and amplitude balance of these components.

The ring hybrid outputs an IQ phase difference of 88-89 degrees, while its quadrature amplitude is 0.5 dB less than the in-phase output. This yields the following phasors (symbol “11” used for analysis purposes):

$$1 + 10^{-0.5/20} e^{88\pi/180} \approx 1.399 \angle 42.41^\circ \quad 1 + 10^{-0.5/20} e^{89\pi/180} \approx 1.387 \angle 42.88^\circ$$

The phase range of the transmitter is therefore $42.88 - 42.41 = 0.47$ degrees.

The IQ mixer at the receiver has an IQ phase difference of 89-91 degrees; while its amplitude balance is exactly matched (0 dB) in the typical case. The total “11” phasor at the demodulated output is therefore

$$1 + e^{89\pi/180} \approx 1.426 \angle 44.50^\circ \text{ (maximum phase deficiency)}$$

$$1 + e^{91\pi/180} \approx 1.402 \angle 45.50^\circ \text{ (maximum phase excess)}$$

The receiver will add 1 degree to the phase range, so the total phase range is 1.47 degrees (which is less than 4.5 degrees and meets specifications).

Cost

The total cost of both the transmitter and receiver is estimated to be approximately \$2,219, with the transmitter accounting for \$901 and the receiver for \$1,218 as shown in Tables 3 and 4.

Transmitter				
Component	Price for Each	Count	Price	Typ Case?
LO	48	1	48	
AMP	279	1	279	
Power Splitter	62.5	1	62.5	Yes
Mixer	144	2	288	
Combiner	50	1	50	
BPF	18.3	1	18.3	
LNA1	52	1	52	
LNA2	85	1	85	
BPF	18.3	1	18.3	Yes
Total Cost			901.1	

Table 3. Cost Breakdown for the Designed Transmitter

Receiver				
Component	Price for Each	Count	Price	Typ Case?
BPF	18.3	1	18.3	Yes
LNA	155	1	155	
1st Coupler	25.24	1	25.24	Yes
Pre-Scaler	74.13	1	74.13	
Amp1	2.18	1	2.18	Yes
2nd Coupler	49	1	49	
Amp2	2.18	1	2.18	Yes
Amp3	30	1	30	Yes
Mixer	279	1	279	
BPF	105	1	105	
Amp4	21.64	1	21.64	
6dB Attenuator	13.05	1	13.05	
Coupler	4.01	1	4.01	Yes
Mixer	172	1	172	
Amp5	36.74	1	36.74	
BPF	18.34	1	18.34	
IQ Mixer	172	1	172	
LPF	20.18	2	40.36	Yes
Total Cost	1218.17			

Table 4. Cost Breakdown for the Designed Receiver

Results

Metric	Required	Achieved
Phase Error	Maximum 4.5°	Transmitter: 0.47° Receiver: 1.47°
Transmit Power	11.5-14.5 dBm	12.7 - 14.0 dBm
SNR at Demodulated Output	Minimum 30 dB	56-68 dB
Receiver Noise Figure	Maximum 3.6 dB	3.28 dB
Largest Output Spurious Signal (Transmitter)	Minimum 43 dBc	51.5 - 79.0 dBc
Largest Output Spurious Signal (Receiver)	Minimum 43 dBc	47.3 - 80.4 dBc

Table 5. Comparison Between Required Specifications and Achieved Specifications.

Having completed the system architecture and explained our design decisions, we now present the results in Table 5. Starting with phase error, we achieved a maximum of 0.47 degrees on the transmitter side and 1.47 degrees on the receiver side. The transmit power ranged from 12.7 dBm to 14.0 dBm. The minimum signal-to-noise ratio (SNR) at the output was calculated to be between 56 dB and 68 dB. The receiver noise figure reached a maximum of 3.28 dB, exceeding our target by approximately 0.3 dB. The output and spurious content, primarily resulting from LO-to-IF leakage, were measured to be in the ranges of 51.5 dBc to 79.0 dBc and 47.3 dBc to 80.4 dBc, respectively.

Overall, the results confirm that our project meets all the required performance specifications.

Conclusion

This project taught us how to approach system-level design from start to finish, which is something we could never have fully learned from textbooks alone. It provided invaluable insight into the common challenges encountered during the design process, including part sourcing difficulties and the need to completely adjust system architecture.

During our final presentation to other design teams, we received constructive feedback that highlighted potential improvements to our approach, as well as key concepts we had initially overlooked—such as amplifier saturation characteristics. Additionally, we realized that our tendency to overestimate the gain and loss ranges of various components made it significantly more difficult to meet specifications and identify appropriate real-world parts. Setting overly wide margins for maximum and minimum values introduced unnecessary complexity and reduced design accuracy.

Overall, the success of this project was made possible by the combined effort of our three-person team as we tackled each challenge together. We are especially grateful to Professor Hausman for his guidance and support throughout the process. His insights helped us better understand and appreciate the complexities of real-world RF system design.

Datasheets

A. Transmitter

Local Oscillator

HMC582LP5

<https://www.analog.com/media/en/technical-documentation/data-sheets/hmc529.pdf>

Amplifier

ADPA9007ACGZN

<https://www.analog.com/media/en/technical-documentation/data-sheets/adpa9007.pdf>

90 Degrees Hybrid Ring

<https://markimicrowave.com/products/surface-mount/quad-hybrids/mqs-0518sm/datasheet/>

Mixer

MM1-0115HPSM-2

<https://markimicrowave.com/products/surface-mount/mixers/mm1-0115hpsm-2/datasheet/>

Band Pass Filter

<https://markimicrowave.com/products/surface-mount/filters/mfba-00003psm/datasheet/>

Combiner

MPD-0422FCSP2

<https://markimicrowave.com/products/surface-mount/power-dividers-combiners/mpd-0422fcsp2/datasheet/>

Low Noise Amplifier

Min 15dB gain Amp

<https://www.analog.com/media/en/technical-documentation/data-sheets/hmc-alh482.pdf>

Compressing Amplifier

<https://markimicrowave.com/products/surface-mount/amplifiers/adm-8624psm/datasheet/>

Band Pass Filter

<https://www.minicircuits.com/pdfs/BFCQ-12600+.pdf>

B. Receiver

Band Pass Filter

<https://www.minicircuits.com/pdfs/BFCQ-12600+.pdf>

First Low Noise Amplifier

<https://markimicrowave.com/products/surface-mount/amplifiers/amm-9852psm/datasheet/>

10dB Coupler

<https://www.minicircuits.com/WebStore/dashboard.html?model=EDC10-183%2B>

-----Carrier Recovery Chain-----

Pre-Scaler: \$74.13

<https://www.mouser.com/datasheet/2/609/hmc447-1503837.pdf>

Amplifier Before Coupler

<https://www.minicircuits.com/pdfs/ERA-9SM+.pdf>

9dB Coupler Before 1st Mixer

<https://www.minicircuits.com/pdfs/BDCH-10-63+.pdf>

Main Path - Amp after the coupler

<https://www.minicircuits.com/pdfs/ERA-9SM+.pdf>

Nominal Coupling Path - Amp after the coupler

<https://markimicrowave.com/products/surface-mount/amplifiers/adm-8096psm/datasheet/>

The 1st Mixer: 0-7 GHz Mixer

<https://markimicrowave.com/products/surface-mount/mixers/mt3-0113lcqg-2/datasheet/>

Band Pass Filter

<https://markimicrowave.com/products/surface-mount/filters/mfbp-00104psm/datasheet/>

Amplifier

<https://www.minicircuits.com/pdfs/PMA1-14LN+.pdf>

6dB Attenuator

<https://www.minicircuits.com/WebStore/dashboard.html?model=BAT-6%2B>

7dB Coupler Before the 2nd Mixer

<https://www.minicircuits.com/pdfs/DCW-6-722+.pdf>

The 2nd Mixer: 1.5 - 13GHz Mixer (Operating at 5.4GHz - 7.2GHz)

<https://markimicrowave.com/products/surface-mount/mixers/mt3h-0113lsm-2/datasheet/>

Amplifier

<https://www.minicircuits.com/WebStore/dashboard.html?model=PMA3-34GLN%2B>

Band Pass Filter

<https://www.minicircuits.com/pdfs/BFCQ-12600+.pdf>

IQ-Mixer

<https://markimicrowave.com/products/surface-mount/iq-mixers/mmiq-0520lsm-2/datasheet/>

LPFs After Mixer

<https://www.minicircuits.com/pdfs/LFHK-2250+.pdf>